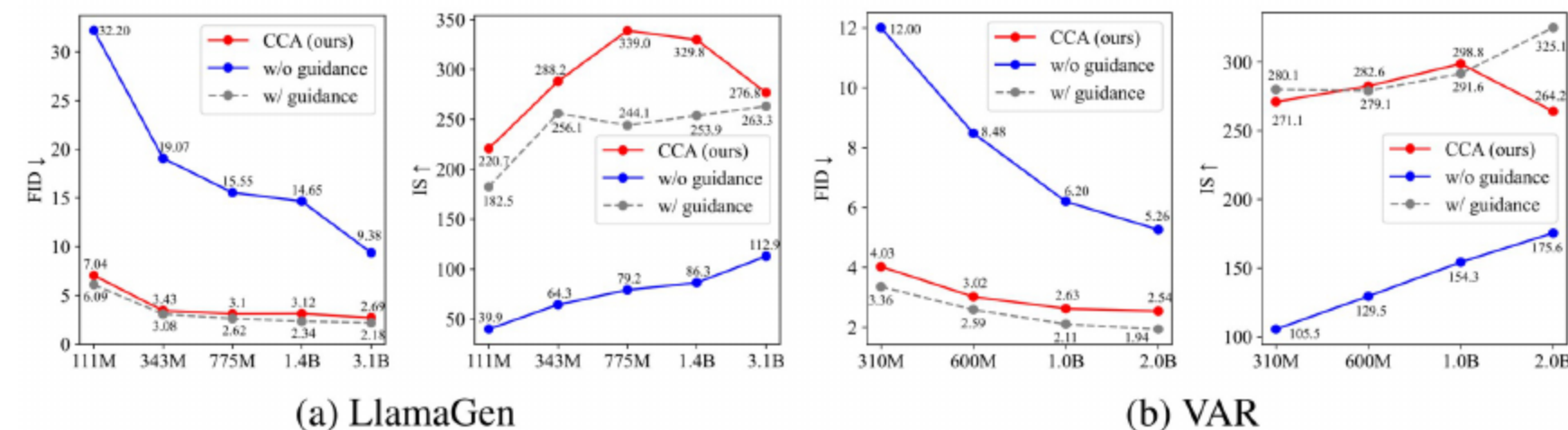


Toward Guidance-Free AR Visual Generation via Condition Contrastive Alignment

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01 Design Gap In AR Visual Generation



- AR visual models rely on sampling-time guidance (CFG), unlike language decoding.
- CFG adds an unconditional branch, doubling per-token inference cost.
- Random condition masking complicates unified multimodal training.

01 Why Guidance-Free Matters

Method	Classifier Guidance	Classifier-Free Guidance	Condition Contrastive Alignment
Modeling of $\log \frac{p(\mathbf{x} \mathbf{c})}{p(\mathbf{x})}$	$\log p_\theta(\mathbf{c} \mathbf{x}) - \log p(\mathbf{c})$	$\log p_\phi(\mathbf{x} \mathbf{c}) - \log p_\theta(\mathbf{x})$	$\beta[\log p_\theta^{\text{sample}}(\mathbf{x} \mathbf{c}) - \log p_\phi(\mathbf{x} \mathbf{c})]$
Training loss	$\max_\theta \mathbb{E}_{p(\mathbf{x},\mathbf{c})} \log p_\theta(\mathbf{c} \mathbf{x})$	$\max_\theta \mathbb{E}_{p(\mathbf{x})} \log p_\theta(\mathbf{x})$	$\min_\theta \mathcal{L}_\theta^{\text{CCA}}$ in Eq. 11
Sampling policy	$\log p_\phi(\mathbf{x} \mathbf{c}) + s \log p_\theta(\mathbf{c} \mathbf{x})$	$(1+s) \log p_\phi(\mathbf{x} \mathbf{c}) - s \log p_\theta(\mathbf{x})$	$\log p_\theta^{\text{sample}}(\mathbf{x} \mathbf{c})$
Extra training cost	$\sim 9\%$ of learning p_ϕ	$\sim 10\%$ of learning p_ϕ	$\sim 1\%$ of pretraining p_ϕ
Sampling cost	$\times \sim 1.3$	$\times 2$	$\times 1$
Applicable area	Diffusion	Diffusion & Autoregressive	Autoregressive

- Unified decoding across text and vision with reduced latency and compute.
- Simpler training by removing unconditional branches and masking.
- Aligns with LLM practice: train-time alignment over sampling-time guidance.

02 Autoregressive Visual Models And Tokenization

- AR models predict next image tokens from vector-quantized discretizations.
- GPT-like training enables unification with language modeling.
- High-fidelity visuals typically need sampling-time guidance.

02 Guided Sampling And Its Target Distribution



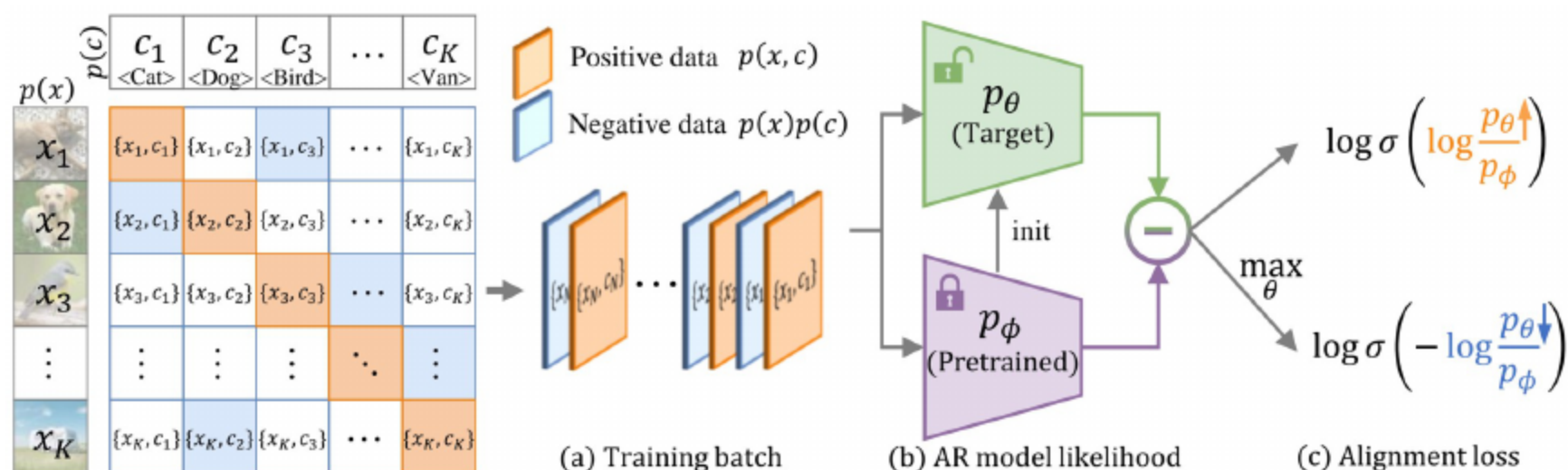
- CFG contrasts conditional and unconditional logits via a scale parameter.
- Classifier and Energy Guidance target similar adjusted distributions.
- Auxiliary models and custom sampling add overhead.

02 LLM Alignment And Direct Preference Optimization

Model	FID↓	IS	sFID↓	Precision	Recall	Model	FID↓	IS	sFID↓	Precision	Recall
LlamaGen-L	19.00	64.7	8.78	0.61	0.67	VAR-d24	6.20	154.3	8.50	0.74	0.62
+DPO	61.69	30.8	44.98	0.36	0.40	+DPO	7.53	232.6	19.10	0.85	0.34
+Unlearn	12.22	111.6	7.99	0.66	0.64	+Unlearn	5.55	165.9	8.41	0.75	0.61
+CCA	3.43	288.2	7.44	0.81	0.52	+CCA	2.63	298.8	7.63	0.84	0.55

- LLMs improve adherence via preference-based fine-tuning.
- DPO optimizes relative likelihoods versus a reference model.
- Motivates training-time alignment to replace guidance.

03 Condition Contrastive Alignment (CCA)



- CCA learns the conditional residual using matched/mismatched pairs.
- Yields a single model approximating guidance targets without changing decoding.
- Lightweight fine-tuning, typically one epoch.

03 Theory Unifying Alignment And Guidance

- Guidance and alignment target the conditional residual $\log p(x|c) - \log p(x)$.
- CCA parameterizes residual as difference between fine-tuned and frozen models.
- Connects visual guidance with language alignment frameworks.

03 Practical Impact



- Guidance-free quality approaches CFG while halving inference cost.
- Preserves standard AR decoding and existing tokenizers.
- Can complement CFG for further gains when acceptable.

04 Target Distribution And Conditional Residual

- Target distribution scales conditional residual between $p(x|c)$ and $p(x)$.
- Log-ratio equals s times residual up to a constant.
- Learning the residual enables single-model approximation of guidance.



04 Method Overview And Theory

Noise Contrastive Estimation Insight

Train $r\theta(x,c)$ via NCE on positives vs. mismatched negatives.

Recovers conditional residual $\log p(x|c) - \log p(x)$.

Uses original pretraining data with shuffled conditions.

Parameterization With A Reference Model

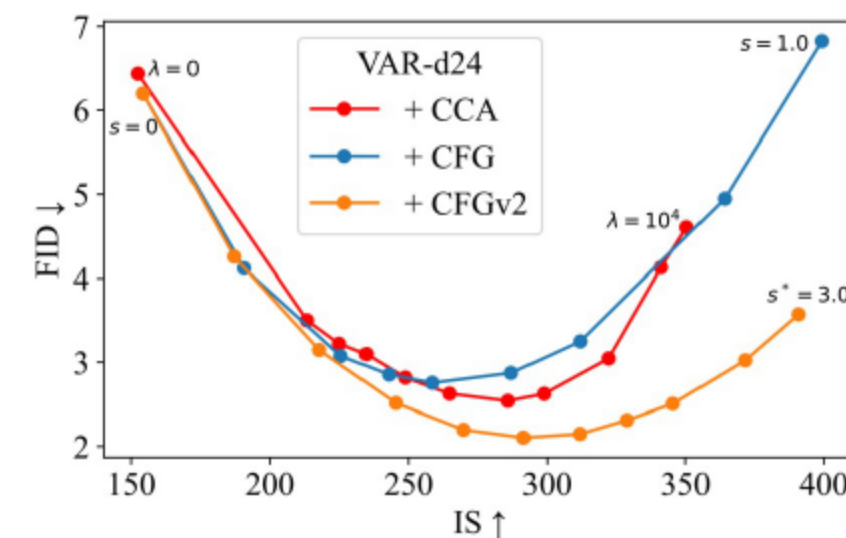
Set $r\theta = \beta[\log p_{\text{sample}\theta}(x|c) - \log p_{\phi}(x|c)]$, p_{ϕ} frozen.

Initialize $p_{\text{sample}\theta}$ from p_{ϕ} and optimize CCA loss.

Inference uses unchanged AR decoding.

05 Batch Construction And Loss

- Shuffle conditions to create negatives from matched pairs.
- Logistic objective contrasts positives vs. negatives relative to reference.
- β controls scale; λ balances fidelity-diversity.



05 Implementation Simplicity

Type	LlamaGen					VAR			
	B 111M	L 343M	XL 775M	XXL 1.4B	3B 3.1B	d16 310M	d20 600M	d24 1.0B	d30 2.0B
CCA β	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02
CCA λ	1000	300	1000	1000	500	50	50	100	1000
CCA+CFG β	0.1	0.02	0.1	0.1	0.1	-	-	-	-
CCA+CFG λ	1	1	1	1	1	-	-	-	-
Learning rate	1e-5	1e-5	1e-5	1e-5	1e-5	2e-5	2e-5	2e-5	2e-5
Dropout?	Yes	Yes	Yes	Yes	Yes	None	Yes	Yes	Yes
Batch size	256	256	256	256	256	256	256	256	256

- Uses only pretraining data; no new annotations.
- Fine-tune ~1 epoch with frozen tokenizer.
- Single-model inference removes unconditional passes and heuristics.

06 Modeling The Conditional Residual



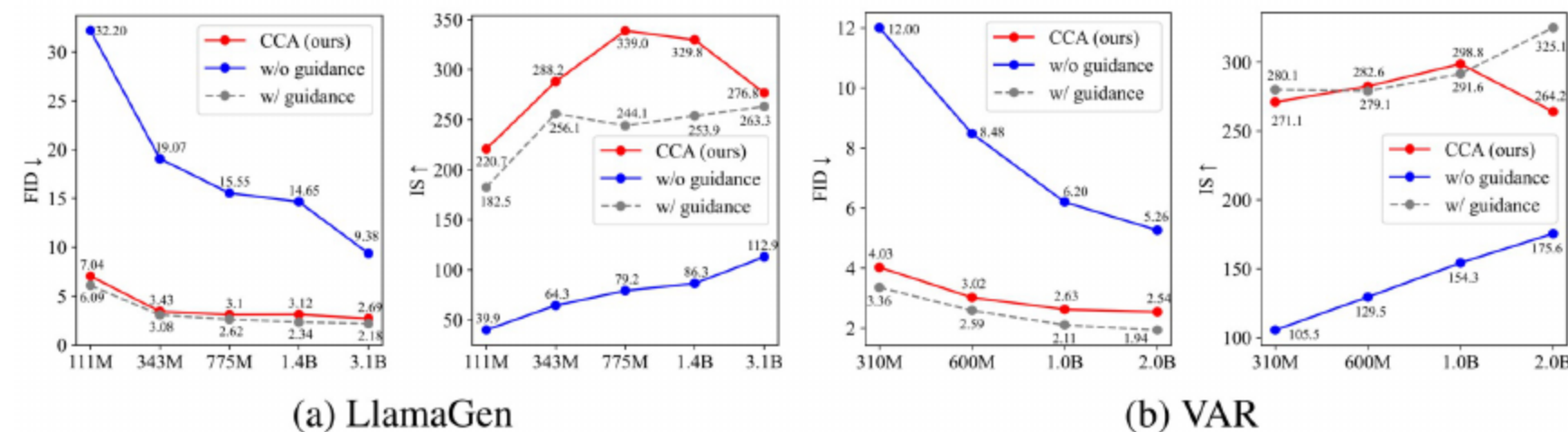
- Classifier Guidance uses $p_{\theta}(c|x)$; CFG uses $p_{\theta}(x)$.
- CCA models $\beta[\log p_{\text{sample}\theta} - \log p_{\phi}]$, avoiding auxiliary networks.
- All target the same sampling distribution.

06 Cost And Policy Differences

Method	Classifier Guidance	Classifier-Free Guidance	Condition Contrastive Alignment
Modeling of $\log \frac{p(x c)}{p(x)}$	$\log p_{\theta}(c x) - \log p(c)$	$\log p_{\phi}(x c) - \log p_{\theta}(x)$	$\beta[\log p_{\theta}^{\text{sample}}(x c) - \log p_{\phi}(x c)]$
Training loss	$\max_{\theta} \mathbb{E}_{p(x,c)} \log p_{\theta}(c x)$	$\max_{\theta} \mathbb{E}_{p(x)} \log p_{\theta}(x)$	$\min_{\theta} \mathcal{L}_{\theta}^{\text{CCA}}$ in Eq. 11
Sampling policy	$\log p_{\phi}(x c) + s \log p_{\theta}(c x)$	$(1+s) \log p_{\phi}(x c) - s \log p_{\theta}(x)$	$\log p_{\theta}^{\text{sample}}(x c)$
Extra training cost	~9% of learning p_{ϕ}	~10% of learning p_{ϕ}	~1% of pretraining p_{ϕ}
Sampling cost	$\times \sim 1.3$	$\times 2$	$\times 1$
Applicable area	Diffusion	Diffusion & Autoregressive	Autoregressive

- Guidance adds training and increases sampling cost.
- CCA needs ~1% extra training and no sampling overhead.
- Sampling policy remains standard AR decoding.

07 Models And Datasets



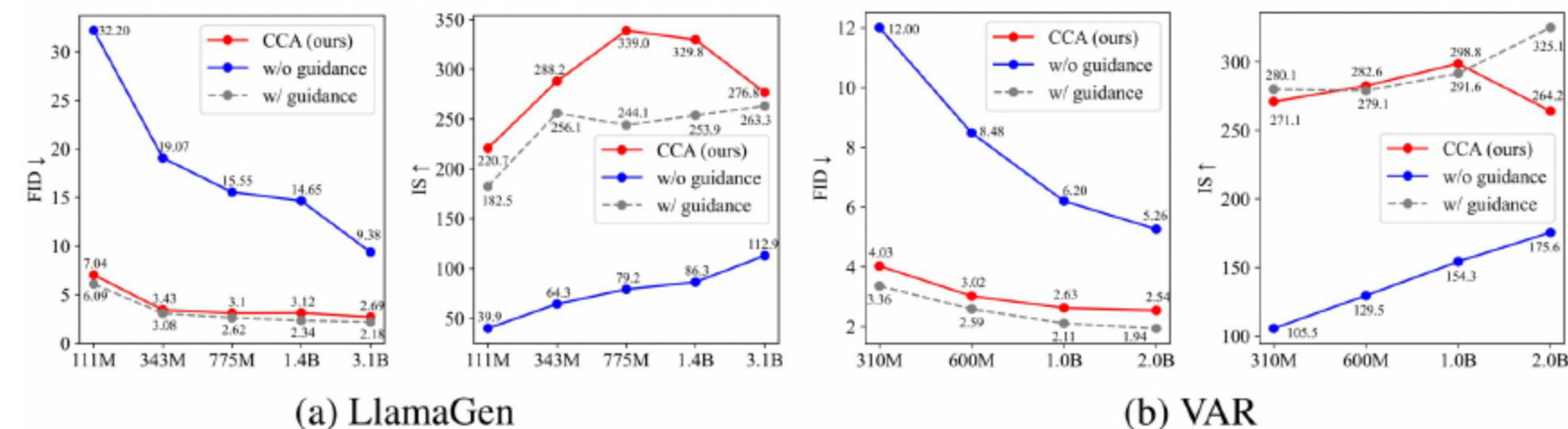
- Evaluate on ImageNet with LlamaGen and VAR families.
- Fine-tune at 384×384 (LlamaGen, eval 256) and 256×256 (VAR).
- Typically one epoch using original setups.

07 Training Hyperparameters

Type	LlamaGen					VAR			
Model Size	B 111M	L 343M	XL 775M	XXL 1.4B	3B 3.1B	d16 310M	d20 600M	d24 1.0B	d30 2.0B
CCA β	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02
CCA λ	1000	300	1000	1000	500	50	50	100	1000
CCA+CFG β	0.1	0.02	0.1	0.1	0.1	-	-	-	-
CCA+CFG λ	1	1	1	1	1	-	-	-	-
Learning rate	1e-5	1e-5	1e-5	1e-5	1e-5	2e-5	2e-5	2e-5	2e-5
Dropout?	Yes	Yes	Yes	Yes	Yes	None	Yes	Yes	Yes
Batch size	256	256	256	256	256	256	256	256	256

- Fix $\beta=0.02$; tune λ per model for fidelity-diversity.
- psample θ initialized from frozen reference $p\phi$.
- Evaluate with FID, IS, precision/recall, sFID.

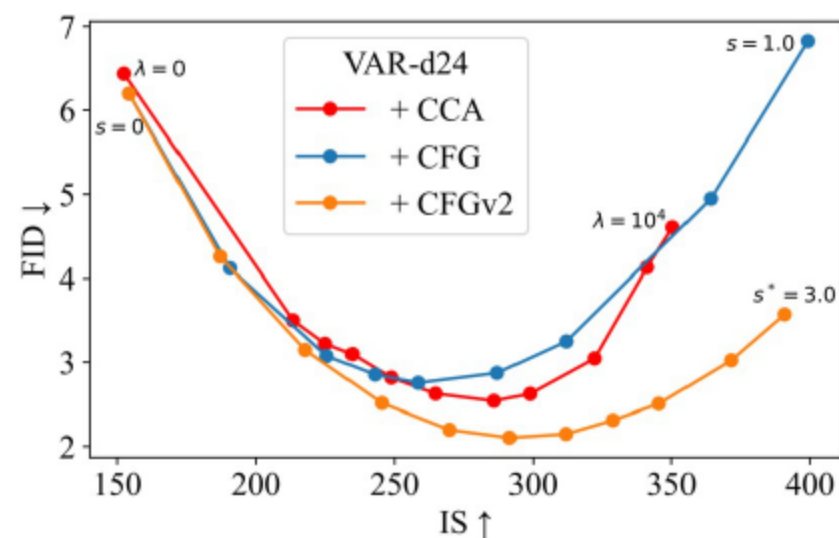
08 Guidance-Free Performance Gains



- CCA boosts guidance-free FID/IS near CFG with one-epoch tuning.
- Improved class alignment and fidelity without custom sampling.

08 Trade-Offs Mirroring CFG

- Varying λ traces FID-IS curves similar to CFG scale.
- Supports equivalence of target distributions.
- λ is practical for balancing diversity and fidelity.



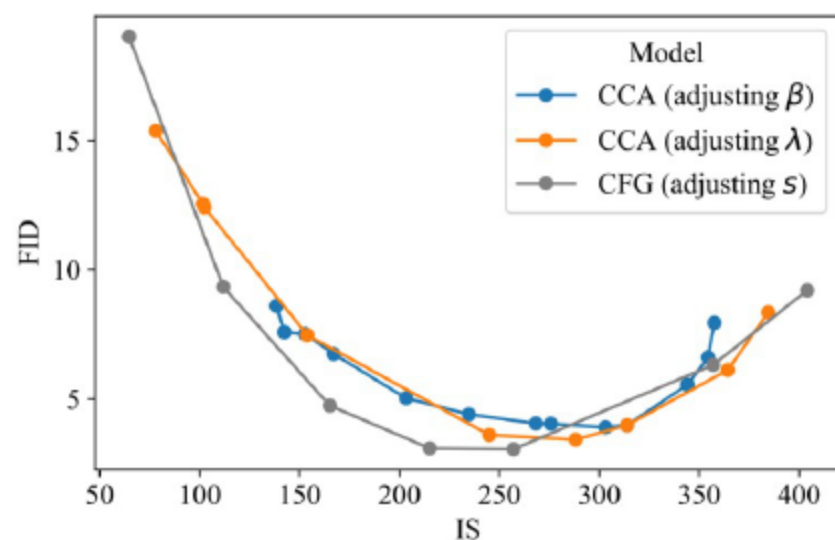
08 Comparison To LLM Alignment Methods

Model	FID↓	IS	sFID↓	Precision	Recall	Model	FID↓	IS	sFID↓	Precision	Recall
LlamaGen-L	19.00	64.7	8.78	0.61	0.67	VAR-d24	6.20	154.3	8.50	0.74	0.62
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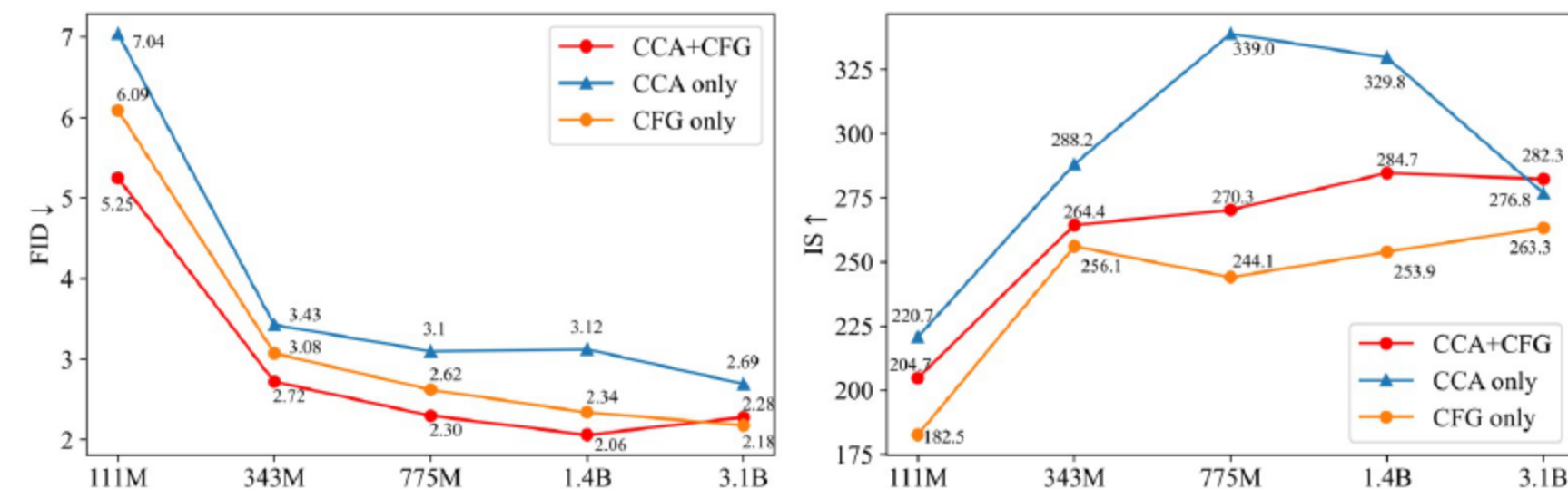
- Direct DPO degrades or collapses in this setting.
- Unlearning yields minor gains, below CCA.
- Frozen reference and contrastive residual stabilize training.

09 Hyperparameter Effects

- λ dominates fidelity-diversity; optima differ for CCA-only vs. CCA+CFG.
- β influences results but is less stable; often fixed.
- Adding CFG helps mainly when λ is reduced.



09 Combining CCA With CFG



- CCA+CFG surpasses CFG when guidance cost is acceptable.
- Maintain occasional condition masking for unconditional branch.
- Optimal λ differs from CCA-only regime.

10 Toward Unified, Efficient AR Visual Generation



- CCA learns conditional residuals, achieving near-CFG quality with half cost.
- Unifies alignment and guidance under a common target distribution.
- Future: broader multimodality, scaling, preference-aware extensions.

THANKS!